AI for Science: Bridging Machine Learning and Scientific Discovery

Pipi Hu

Beijing Institute of Mathematical Sciences and Applications

October 7, 2025

Feynman's words

What I cannot create, I do not understand.

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

What is AI for Science?

AI for Science represents the intersection of artificial intelligence and scientific research, where machine learning methods are applied to accelerate scientific discovery.

- Drug Discovery: Predicting molecular properties and designing new compounds
- Materials Science: Discovering new materials with desired properties
- Climate Modeling: Understanding and predicting climate patterns
- Physics: Solving complex physical systems and equations
- Biology: Understanding protein folding and biological processes

Key Challenges in AI for Science

Scientific problems present unique challenges for machine learning:

1. **Data scarcity**: Limited experimental data, expensive to collect
2. **Physical constraints**: Solutions must obey physical laws
3. **Interpretability**: Scientific understanding requires explainable models
4. **Uncertainty quantification**: Scientific predictions need confidence bounds
5. **Multi-scale phenomena**: From quantum to macroscopic scales

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Physics-Informed Neural Networks

PINNs incorporate physical laws directly into neural network training:

$$\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{physics} \tag{1}$$

where

- $\mathcal{L}_{data}$: Standard data fitting loss
- $\mathcal{L}_{physics}$: Physics constraint loss (PDEs, conservation laws)
- λ: Weighting parameter balancing data and physics

Example: Solving PDEs with PINNs

Consider the heat equation:

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u + f(x, t) \tag{2}$$

with boundary conditions $u(x, 0) = g(x)$ and $u(\partial\Omega, t) = h(x, t)$.
The physics loss becomes:

$$\mathcal{L}_{physics} = \left\| \frac{\partial u_{NN}}{\partial t} - \alpha \nabla^2 u_{NN} - f \right\|^2 \tag{3}$$

where $u_{NN}(x, t; \theta)$ is the neural network approximation.

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Advantages of PINNs

- **No mesh required**: Unlike traditional numerical methods
- **Unified framework**: Handle forward and inverse problems
- **Data fusion**: Combine sparse data with physical knowledge
- **Parameter discovery**: Learn unknown parameters in PDEs

Challenges:

- Training can be unstable
- Requires careful hyperparameter tuning
- May struggle with complex geometries

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Molecular Representation Learning

Converting molecules to machine-readable formats:

- **SMILES strings**: Text-based molecular representation
- **Graph neural networks**: Molecular graphs with atoms as nodes, bonds as edges
- **3D coordinates**: Spatial arrangement of atoms
- **Descriptors**: Hand-crafted molecular features

Graph Neural Networks for Molecules

For a molecular graph $G = (V, E)$ with atoms $v_i \in V$ and bonds $e_{ij} \in E$:
Message passing:

$$h_v^{(l+1)} = \text{UPDATE}\left(h_v^{(l)}, \text{AGGREGATE}\left(\{h_u^{(l)} : u \in \mathcal{N}(v)\}\right)\right) \tag{4}$$

Property prediction:

$$y = \text{READOUT}(\{h_v^{(L)} : v \in V\}) \tag{5}$$

where L is the number of message passing layers.

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Applications in Drug Discovery

- **Property prediction**: Solubility, toxicity, binding affinity
- **Molecular generation**: Designing new drug candidates
- **Retrosynthesis**: Planning synthetic routes
- **Protein-ligand binding**: Predicting drug-target interactions

Key datasets:

- QM9: 134k small organic molecules
- ChEMBL: Large-scale bioactivity data
- PDB: Protein structures

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Climate Modeling with AI

Climate systems involve complex interactions across multiple scales:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F} \tag{6}$$

where $\mathbf{u}$ is velocity, p is pressure, and $\mathbf{F}$ represents external forces.

Weather Prediction

Traditional approach: Numerical weather prediction (NWP) using physics-based models.
AI approach: Deep learning models trained on historical weather data:

- **ConvLSTM**: Spatiotemporal weather prediction
- **Graph neural networks**: Modeling atmospheric dynamics
- **Transformer models**: Long-range dependencies in weather patterns

Hybrid approach: Combining AI with traditional NWP for improved accuracy.

Climate Change Detection

Using AI to detect and attribute climate change:

- **Pattern recognition**: Identifying climate change signatures
- **Attribution studies**: Linking extreme events to climate change
- **Scenario modeling**: Predicting future climate under different emissions

Key challenges:

- Limited historical data
- Non-stationary climate system
- Uncertainty quantification

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

The Protein Folding Problem

Proteins fold from linear amino acid sequences to 3D structures:

$$\text{Sequence: ACDEFGHIKLMNPQRSTVWY} \rightarrow \text{3D Structure} \tag{7}$$

Key challenges:

- Exponential number of possible conformations
- Complex energy landscapes
- Multiple timescales (nanoseconds to seconds)

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

AlphaFold approach:

1. **Multiple sequence alignment**: Evolutionary information
2. **Attention mechanisms**: Long-range interactions
3. **Structure module**: 3D coordinate prediction
4. **Confidence estimation**: Reliability of predictions

Key innovations:

- End-to-end differentiable architecture
- Integration of evolutionary and physical constraints
- Uncertainty quantification

Impact of AlphaFold

- **Structural biology revolution**: Millions of protein structures predicted
- **Drug discovery**: Better understanding of drug targets
- **Protein design**: Engineering proteins with desired functions
- **Evolutionary biology**: Understanding protein evolution

Current limitations:

- Limited accuracy for disordered regions
- Challenges with protein complexes
- Difficulty with membrane proteins

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Materials Property Prediction

Predicting material properties from composition and structure:

- **Electronic properties**: Band gap, conductivity, magnetic properties
- **Mechanical properties**: Elastic modulus, strength, hardness
- **Thermal properties**: Heat capacity, thermal conductivity
- **Optical properties**: Refractive index, absorption spectra

Crystal Structure Representation

Periodic systems: Representing crystal structures with periodic boundary conditions.

Graph representations:

- **Crystal graphs**: Atoms as nodes, bonds as edges
- **Periodic embeddings**: Handling translational symmetry
- **Multi-scale features**: From atomic to macroscopic properties

Deep learning approaches:

- **CGCNN**: Crystal Graph Convolutional Neural Networks
- **SchNet**: Quantum-chemical property prediction
- **MEGNet**: Materials property prediction

High-Throughput Materials Discovery

Computational screening:

1. Generate candidate materials
2. Predict properties using ML models
3. Filter promising candidates
4. Experimental validation

Key databases:

- **Materials Project**: 150k+ materials with computed properties
- **OQMD**: Open Quantum Materials Database
- **AFLOW**: Automatic Flow for Materials Discovery

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Accelerating Scientific Simulations

AI can accelerate traditional scientific computing:

- **Surrogate models**: Fast approximations of expensive simulations
- **Multiscale modeling**: Bridging different length and time scales
- **Parameter optimization**: Finding optimal simulation parameters
- **Uncertainty quantification**: Propagating uncertainties through simulations

Neural Operators

Learning mappings between function spaces:

$$G : \mathcal{U} \to \mathcal{V}, \quad u \mapsto v \tag{8}$$

where $\mathcal{U}$ and $\mathcal{V}$ are function spaces.

Examples:

- **Neural ODEs**: Learning dynamics of ordinary differential equations
- **Neural PDEs**: Solving partial differential equations
- **Fourier Neural Operators**: Learning in frequency domain

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Applications in Scientific Computing

- **Fluid dynamics**: Turbulence modeling, flow prediction
- **Quantum chemistry**: Electronic structure calculations
- **Solid mechanics**: Stress analysis, fracture prediction
- **Electromagnetics**: Wave propagation, antenna design

Benefits:

- Orders of magnitude speedup
- Reduced computational cost
- Real-time predictions

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Current Challenges in AI for Science

1. **Data quality and quantity**: Limited, noisy, or biased scientific data
2. **Physical consistency**: Ensuring AI predictions obey physical laws
3. **Interpretability**: Understanding what AI models learn
4. **Generalization**: Models that work across different domains
5. **Uncertainty quantification**: Reliable error estimates

Future Directions

- **Foundation models for science**: Large-scale models trained on scientific data
- **Automated hypothesis generation**: AI systems that propose new scientific theories
- **Scientific reasoning**: AI that can perform logical scientific reasoning
- **Multimodal scientific AI**: Integrating different types of scientific data
- **Scientific collaboration**: AI systems that work alongside human scientists

Ethical Considerations

- **Scientific integrity**: Ensuring AI doesn't introduce bias in scientific research
- **Reproducibility**: Making AI for science research reproducible
- **Open science**: Sharing models, data, and code
- **Scientific communication**: How AI affects scientific publishing and peer review

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Case Study 1: Drug Discovery

Problem: Discovering new drugs for COVID-19 treatment.

AI approach:

- Molecular property prediction
- Virtual screening of compound libraries
- Drug-target interaction prediction
- Synthesis route planning

Results: Several AI-discovered compounds entered clinical trials.

北京雁栖湖
应用数学研究院
Beijing Institute
of Mathematical
Sciences and Applications

Case Study 2: Climate Modeling

Problem: Improving climate change predictions.

AI approach:

- Hybrid physics-AI models
- Downscaling global climate models
- Extreme weather event prediction
- Climate change attribution

Impact: Better understanding of climate change impacts and adaptation strategies.

Case Study 3: Materials Discovery

Problem: Finding new materials for renewable energy applications.

AI approach:

- High-throughput computational screening
- Property prediction models
- Structure-property relationships
- Synthesis pathway optimization

Results: Discovery of new photovoltaic and battery materials.

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Summary

AI for Science represents a paradigm shift in how we approach scientific research:

- **Acceleration**: Faster discovery and development cycles
- **Integration**: Combining data-driven and physics-based approaches
- **Automation**: Reducing manual work in scientific research
- **Insights**: Discovering patterns humans might miss

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

The Future of AI for Science

- **Democratization**: Making advanced scientific tools accessible to all researchers
- **Interdisciplinary collaboration**: Breaking down silos between scientific fields
- **Scientific method evolution**: How AI changes the scientific process
- **Grand challenges**: Tackling humanity's biggest scientific problems

Questions for Discussion

1. How can we ensure AI for Science maintains scientific rigor and reproducibility?
2. What are the implications of AI for the future of scientific careers?
3. How should we balance AI automation with human scientific intuition?
4. What ethical guidelines should govern AI for Science research?

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Welcome inputs

Any comments?

Welcome your inputs!

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications